

# Schmidt Sciences · Trustworthy AI Research Agenda

Annotated with Mechanistic Views mappings | 38 questions | [mechanistic-views.github.io](https://mechanistic-views.github.io)

Dissolved · question was ill-posed, disappears when you name the view

Reframed · framework clarifies what evidence would answer it

Scoped · framework identifies which view owns it

Outside scope · methods-level or multi-agent

## Characterizing and Forecasting Misalignment

### Q1.1.1

What does it mean, in decision-relevant terms, for an AI system to be misaligned, and how can we quantify or bound misalignment?

-> OUTSIDE: Methods-level: operationalizing a specific target property

### Q1.1.2

Under what conditions do models exploit training objective flaws or pursue unintended goals (specification gaming, goal misgeneralization)?

-> OUTSIDE: Methods-level: detecting specific failure modes

### Q1.1.3

Which environmental changes increase misalignment risk, and what causes behaviors to emerge that weren't apparent during training?

-> SCOPED [Training dynamics]: Training dynamics / Process view territory

### Q1.1.4

How do extended human-model interactions shape behavior? When do models develop manipulative or approval-seeking dynamics?

-> OUTSIDE: Multi-agent scope boundary

## Mechanisms of Generalization and Representation

### Q1.2.5

Why do models converge on particular solutions? How do architecture, optimization, data, and scale shape learned representations?

-> SCOPED [Training dynamics]: Training dynamics · Process view territory

### Q1.2.6

When do models behave as if possessing internal goal representations? How do these emerge and relate to misalignment?

-> DISSOLVED [Probe features]: "Behave as if" = Instrumental view. "Possess" = Subspace or Structural. Different views, different answers.

### Q1.2.7

When do representations support causal reasoning? Do richer world models improve robustness or enable constraint evasion?

-> DISSOLVED [Linear assumption]: Linear encodings = Subspace claim. Causal world model = Structural claim. Two views conflated.

### Q1.2.8

When does training compress intended objectives into proxies that misgeneralize? Can we detect such proxies internally?

-> SCOPED [Validation methodology]: Validation methodology · detecting artifacts requires discriminant validity (V3)

## Scaling, Emergence, and Forecasting Risk

### Q1.3.9

How do autonomy, effective time horizon, and capability uplift scale with model size and compute?

-> OUTSIDE: Methods-level: scaling laws are empirical, not framework

### Q1.3.10

When do safety-relevant capabilities emerge during training? Are there predictable phase transitions?

-> SCOPED [Training dynamics]: Training dynamics · Process view. Phase transitions = E5 Temporal stability.

### Q1.3.11

Which observable signals best predict future misbehavior under deployment-like conditions?

-> OUTSIDE: Methods-level: empirical prediction problem

### Q1.3.12

What structured arguments justify relying on evaluations outside tested settings?

-> SCOPED [Validation methodology]: Validation methodology · E2 Distribution shift, I4 Confound control

## Building a Science of Evaluation

### Q2.1.13

How can evaluations measure safety-relevant constructs with defensible evidence and auditable behavioral indicators?

-> SCOPED [Validation methodology]: Uses MV vocabulary directly · 'construct validity' = our C1-C5 criteria

### Q2.1.14

What evidence demonstrates evaluation performance in real-world settings?

-> SCOPED [Validation methodology]: External validity · E1 Task transfer, E2 Distribution shift

### Q2.1.15

When an evaluation or detector appears to generalize, what structured evidence is sufficient to justify relying on it in a safety case?

-> SCOPED [Validation methodology]: Validation methodology · I4 Confound, E2 Distribution. Framework provides the evidence standards.

### Q2.1.16

How can evaluations identify rare, delayed, or trajectory-dependent behavior?

-> OUTSIDE: Methods-level: engineering rare-event detection

### Q2.1.17

How do we build evaluations remaining valid when explicitly optimized against?

-> SCOPED [Safety evidence gaps]: Structural view · gauge-invariant structure survives adversarial pressure

### Q2.1.18

Can simplified settings reliably elicit misaligned behavior for systematic measurement?

-> SCOPED [Validation methodology]: Validation methodology · V3 Discriminant validity in controlled settings

### Q2.1.19

Can we estimate probability of consequential but low-frequency failures?

-> OUTSIDE: Methods-level: statistical estimation problem

### Q2.1.20

When are intermediate outputs (reasoning traces) informative about actual model reasoning versus strategically optimized appearance?

-> REFRAMED [CoT faithfulness]: = CoT faithfulness. 'Faithful' means 4 things. Perspectival view predicts single-method unreliability.

## Interventions that Generalize

### Q2.2.21

Which interventions change underlying learned drivers rather than surface behavior? When do mechanistic interventions offer advantages?

-> SCOPED [Intervention limits]: Intervention limits · Object ceiling. Ablation may break, not reveal.

### Q2.2.22

When do alignment methods like deliberative training improve robustness under pressure?

-> OUTSIDE: Methods-level: evaluating specific training techniques

### Q2.2.23

How far can improved specifications reduce misalignment? How can models represent normative ambiguity?

-> OUTSIDE: Methods-level: specification engineering

### Q2.2.24

Are there interventions differentially preserving human agency rather than substituting for it?

-> OUTSIDE: Sociotechnical / governance · outside framework scope

## Oversight as Training Signal

### Q3.1a.25

How can amplified oversight achieve sufficiently reliable coverage to avoid specification gaming?

-> OUTSIDE: Methods-level: oversight engineering

### Q3.1a.26

How do human biases interact with amplified oversight? What designs resist correlated evaluator errors?

-> OUTSIDE: Methods-level: human factors in evaluation

### Q3.1a.27

When do amplified oversight methods constrain learned goals? How can we elicit honest internal information?

-> DISSOLVED [CoT faithfulness]: 'Honest internal information' = view-dependent. Instrumental: predicts behavior. Structural: reflects computation. Different evidence needed.

### Q3.1a.28

Can we learn from partial reasoning supervision, bridging supervised learning and preference learning?

-> OUTSIDE: Methods-level: training methodology

### Q3.1a.29

Can amplified oversight identify misspecified training objectives?

-> OUTSIDE: Methods-level: oversight engineering

## Oversight as Deployment-Time Control

### Q3.1b.30

What properties about stronger models can weaker overseers reliably verify or bound?

-> OUTSIDE: Methods-level: scalable oversight problem

### Q3.1b.31

How can we verify whether oversight methods generalize beyond tested failure modes?

-> SCOPED [Validation methodology]: Validation methodology · E2 Distribution shift

### Q3.1b.32

How can oversight remain effective when models condition behavior on monitoring? How do we detect strategies with delayed effects?

-> SCOPED [Safety evidence gaps]: Safety evidence gaps · Instrumental evidence insufficient for Structural claims about strategic behavior

## Multi-Agent Risks

### Q3.2.33

How do we measure coerciveness, strategic competence, positional preferences, and partiality emerging through interaction?

-> OUTSIDE: Multi-agent scope boundary · genuine limitation of single-model framework

### Q3.2.34

How do specification gaming, proxy collapse, and distribution shift manifest when multiple agents interact?

-> OUTSIDE: Multi-agent scope boundary

### Q3.2.35

When do agents coordinate to defeat oversight? How can evaluations detect collusion?

-> OUTSIDE: Multi-agent scope boundary

### Q3.2.36

What failures emerge at collective levels? How do local failures propagate through agent networks?

-> OUTSIDE: Multi-agent scope boundary

**Q3.2.37**

What technical tooling enables trustworthy multi-agent interaction?

-> OUTSIDE: Multi-agent scope boundary

**Q3.2.38**

What risks unfold over extended periods in multi-agent settings (competitive drift, norm erosion, lock-in)?

-> OUTSIDE: Multi-agent scope boundary